Digital Filters

Physical Measuring Techniques FK7063

Michael Vogt
June 2026

Introduction

In this computational lab, we construct several digital filters using Matlab and compare them to each other. We also analyze data from an EKG signal and determine ways to remove noise from it.

1 Tasks

1.1 Constructing a Second Order Stop-Band Filter

We define the frequency response of a stop-band filter whose gain is 0 at $f = f_0 = 50\,\mathrm{Hz}$ with a bandwidth of $b = 4\,\mathrm{Hz}$ at $-3\,\mathrm{dB}$ (corresponding to an attenuation of $1/\sqrt(2)$). The sampling rate is $F_s = 1000\,\mathrm{Hz}$.

In the space of the z-transform, the frequency f_0 corresponds to $\exp(i\Omega_0)$ with

$$\Omega_0 = 2\pi\frac{f_0}{F_s} = \frac{\pi}{10}$$

To get the desired attenuation, we place a zero at this point, i. e. we add a factor $(1-\exp(i\Omega_0))$ to the denominator. Since real signals always contain negative frequencies in the same amount as corresponding positive frequencies, we also need to attenuate $-50\,\mathrm{Hz}$ (corresponding to $-\Omega_0$) in the same way, so we add a factor $(1 - \exp(-i\Omega 0))$.

Now we need to add the property that the filter's maximum gain is 1 and that it has the desired bandwidth. This is achieved by adding a pole (i.e. a factor in the denominator) that is not on the unit circle but still close to the zero that we added. It can be parametrized as $r\exp(i\Omega 0)$ (with r real). For values of z that are far away, the distance to the pole and the zero will be roughly the same, so modulus of the numerator and denominator are equal, leading to a gain of 1. When z approaches the pole and the zero, there comes a point when it is separated from the zero by a distance $d \approx (1-r)$ and from the pole by a distance $\approx \sqrt{d + (1-r)} \approx \sqrt{2}(1-r)$. the ratio of the numerator and denominator is then $1/\sqrt(2)$, so the arc around the unit circle where the gain is lower

Figure 1

than $-3\,\mathrm{dB}$ has a length of $2d \approx 2(1-r)$. The bandwidth of the filter thus depends on the distance r of the pole to the origin. We require the arc $2d$ to correspond to the bandwidth $b = 4\,\mathrm{Hz}$:

$$2(1-r) = 2d \stackrel{!}{=} 2\pi \frac{b}{F_s}$$
$$\Longrightarrow r = 1 - \frac{b}{F_s}\pi = 1 - \frac{\pi}{250} = 0.98743$$

Again, we at two poles for positive and negative frequencies. With both zeros and poles, the filter transfer function becomes

$$H(z) = \frac{(1-\exp(i\Omega_0))(1-\exp(-i\Omega_0))}{(1-r\exp(i\Omega_0))(1-r\exp(-i\Omega_0))}$$

Multiplying out the brackets and then diving through by z^2, we get $H(z)$ as a ratio of polynomials in z^{-1}:

$$\begin{aligned} H(z) = \frac{B(z)}{A(z)} &= \frac{1-2\cos(\Omega_0)\cdot z^{-1} + z^{-2}}{1-2r\cos(\Omega_0)\cdot z^{-1} + r^2 \cdot z^{-2}} \\ &= \frac{1-1.9021z^{-1}+z^{-2}}{1-1.8782z^{-1}+0.9750z^{-2}} \end{aligned}$$

This is the form used by the Matlab function `freqz(b, a, N)` to plot the transfer function. b and a specify the coefficients in $B(z)$ and $A(z)$ so `b = [1 -1.9021 1]` and `a = [1 -1.8782 0.9750]`. This gives the plot shown in Fig. **??**

2 Conclusion